

## PRINCETON, MN, USA

WMO: 726682

Lat: 45.560N

Lon: 93.608W

Elev: 298

StdP: 97.79

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54914

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -27.1      | -23.0 | -30.9                         | 0.2 | -25.6 | -27.6 | 0.3 | -22.7 | 9.4                   | -8.5 | 8.6 | -8.0 | 1.6                   | 280  | 0.554 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.1                   | 32.2            | 22.5 | 30.2 | 21.6 | 28.7 | 20.8 | 24.3                | 29.3 | 23.2 | 28.3 | 22.1 | 27.0 | 4.6                  | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 22.6                            | 17.9 | 27.7 | 22.0 | 17.2 | 27.0 | 20.9 | 16.1 | 25.7 | 74.9          | 29.4 | 70.4 | 28.0 | 66.2 | 27.0 | 28.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (d)                        | (e)   | (f)                | (g) | (h)                                                | (i)   | (j)        | (k)   | (l)        | (m)   | (n)        | (o)   | (p)  | (q) |
| (4) 9.1           | 8.1  | 7.3 | DB                         | -31.4 | 34.5               | 3.1 | 1.5                                                | -33.6 | 35.6       | -35.4 | 36.5       | -37.1 | 37.3       | -39.3 | 38.5 | (4) |
| (5)               |      |     | WB                         | -31.6 | 26.0               | 3.0 | 1.1                                                | -33.7 | 26.8       | -35.5 | 27.4       | -37.2 | 28.1       | -39.4 | 28.9 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 6.7    | -10.2 | -8.2  | -0.9  | 7.2   | 13.6  | 18.9  | 21.7  | 20.0  | 15.9  | 8.2   | 0.5   | -7.1  |
|                                                                        | DBStd       | 12.43  | 7.47  | 7.06  | 7.05  | 5.49  | 4.74  | 3.86  | 3.28  | 3.20  | 4.80  | 5.12  | 5.89  | 6.94  |
|                                                                        | HDD10.0     | 2535   | 626   | 508   | 343   | 117   | 18    | 1     | 0     | 0     | 7     | 96    | 290   | 530   |
|                                                                        | HDD18.3     | 4526   | 884   | 742   | 597   | 337   | 161   | 38    | 8     | 18    | 102   | 315   | 536   | 788   |
|                                                                        | CDD10.0     | 1338   | 0     | 0     | 4     | 32    | 129   | 268   | 363   | 311   | 186   | 41    | 4     | 0     |
|                                                                        | CDD18.3     | 288    | 0     | 0     | 1     | 2     | 14    | 56    | 113   | 71    | 30    | 2     | 0     | 0     |
|                                                                        | CDH23.3     | 2940   | 0     | 0     | 2     | 23    | 205   | 558   | 1153  | 686   | 288   | 24    | 0     | 0     |
|                                                                        | CDH26.7     | 894    | 0     | 0     | 0     | 5     | 64    | 154   | 396   | 196   | 78    | 2     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 2.9    | 2.9   | 3.0   | 3.1   | 3.5   | 3.3   | 2.8   | 2.4   | 2.3   | 2.7   | 3.0   | 3.0   | 2.8   |
| (15) Precipitation                                                     | PrecAvg     | 753    | 20    | 20    | 40    | 69    | 82    | 114   | 101   | 96    | 79    | 70    | 39    | 23    |
|                                                                        | PrecMax     | 960    | 52    | 45    | 85    | 198   | 160   | 211   | 179   | 202   | 162   | 141   | 118   | 56    |
|                                                                        | PrecMin     | 565    | 2     | 2     | 17    | 14    | 25    | 56    | 34    | 14    | 17    | 33    | 0     | 6     |
|                                                                        | PrecStd     | 122    | 13    | 11    | 20    | 39    | 38    | 41    | 38    | 43    | 44    | 34    | 33    | 13    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 7.3   | 11.3  | 21.0  | 26.3  | 32.3  | 32.7  | 34.0  | 32.9  | 32.0  | 26.3  | 18.8  | 8.7   |
|                                                                        |             | MCWB   | 3.5   | 7.8   | 13.6  | 15.2  | 19.6  | 23.1  | 23.4  | 22.8  | 21.8  | 17.3  | 12.1  | 7.1   |
|                                                                        | 2%          | DB     | 3.7   | 7.0   | 16.1  | 22.5  | 27.8  | 30.1  | 32.3  | 31.0  | 28.8  | 22.7  | 15.2  | 5.9   |
|                                                                        |             | MCWB   | 1.6   | 3.9   | 10.9  | 11.8  | 17.4  | 21.1  | 23.2  | 22.4  | 20.8  | 15.5  | 10.4  | 3.2   |
|                                                                        | 5%          | DB     | 2.1   | 3.8   | 12.4  | 19.8  | 25.9  | 28.0  | 30.9  | 28.7  | 27.1  | 19.1  | 12.2  | 2.9   |
|                                                                        |             | MCWB   | 0.9   | 1.9   | 8.5   | 11.2  | 16.1  | 19.4  | 22.5  | 21.3  | 19.6  | 13.5  | 8.3   | 1.5   |
|                                                                        | 10%         | DB     | 0.0   | 2.2   | 8.8   | 17.3  | 22.6  | 27.1  | 28.8  | 27.3  | 24.0  | 17.2  | 8.8   | 2.1   |
|                                                                        |             | MCWB   | -0.8  | 0.7   | 5.8   | 9.7   | 14.5  | 18.9  | 21.5  | 20.0  | 18.1  | 12.1  | 5.9   | 1.0   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 3.7   | 7.8   | 15.5  | 16.5  | 22.1  | 24.5  | 26.1  | 25.1  | 24.1  | 19.5  | 14.2  | 6.7   |
|                                                                        |             | MCDB   | 6.6   | 10.8  | 19.0  | 24.9  | 29.0  | 30.0  | 31.4  | 30.8  | 28.4  | 23.3  | 17.0  | 8.3   |
|                                                                        | 2%          | WB     | 1.9   | 4.1   | 11.6  | 14.0  | 19.4  | 22.7  | 24.7  | 23.7  | 22.1  | 16.7  | 11.3  | 3.3   |
|                                                                        |             | MCDB   | 3.5   | 6.4   | 14.9  | 20.0  | 24.8  | 28.0  | 29.8  | 28.6  | 26.7  | 20.9  | 14.9  | 4.8   |
|                                                                        | 5%          | WB     | 0.5   | 2.1   | 8.6   | 11.9  | 17.6  | 21.4  | 23.6  | 22.4  | 20.7  | 14.7  | 8.2   | 1.9   |
|                                                                        |             | MCDB   | 1.7   | 4.0   | 12.6  | 17.6  | 22.6  | 26.6  | 28.6  | 27.1  | 24.8  | 18.8  | 11.2  | 3.1   |
|                                                                        | 10%         | WB     | -1.0  | 0.4   | 5.6   | 10.2  | 15.8  | 20.0  | 22.5  | 21.1  | 19.3  | 12.5  | 6.3   | 0.6   |
|                                                                        |             | MCDB   | -0.2  | 1.9   | 8.8   | 16.0  | 21.0  | 24.9  | 27.5  | 25.4  | 23.5  | 16.0  | 9.0   | 1.7   |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 9.7   | 10.5  | 10.6  | 11.9  | 12.6  | 11.7  | 12.1  | 12.3  | 12.5  | 11.1  | 9.4   | 8.4   |
|                                                                        | 5% DB       | MCDBR  | 10.1  | 10.5  | 14.2  | 16.6  | 16.9  | 13.7  | 13.6  | 13.9  | 14.3  | 15.3  | 13.8  | 9.3   |
|                                                                        |             | MCWBR  | 8.3   | 7.9   | 9.2   | 8.8   | 8.0   | 6.4   | 6.1   | 6.6   | 7.3   | 8.7   | 9.0   | 7.6   |
|                                                                        | 5% WB       | MCDBR  | 9.6   | 9.8   | 13.4  | 14.0  | 13.9  | 11.7  | 11.7  | 11.8  | 11.9  | 12.5  | 12.5  | 8.4   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.263 | 0.273 | 0.313 | 0.372 | 0.406 | 0.392 | 0.386 | 0.398 | 0.370 | 0.316 | 0.286 | 0.262 |
|                                                                        | taud        |        | 2.397 | 2.366 | 2.375 | 2.292 | 2.249 | 2.333 | 2.373 | 2.341 | 2.392 | 2.538 | 2.508 | 2.425 |
|                                                                        | Ebn at Noon |        | 858   | 918   | 920   | 886   | 863   | 875   | 876   | 850   | 847   | 859   | 827   | 821   |
|                                                                        | Edh at Noon |        | 71    | 89    | 103   | 123   | 133   | 123   | 117   | 116   | 102   | 76    | 65    | 63    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.70  | 2.79  | 3.88  | 4.65  | 5.34  | 5.91  | 6.26  | 5.22  | 4.03  | 2.52  | 1.59  | 1.18  |
|                                                                        | RadStd      |        | 0.19  | 0.27  | 0.34  | 0.43  | 0.43  | 0.43  | 0.21  | 0.39  | 0.28  | 0.34  | 0.14  | 0.15  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (61 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page